



Instructor: Date:

Learner: Regional Branch:

Program: Center:

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Let's Start!

The Key Facts of Operations on Variables and Values

Learning Outcome:

Understand information about web applications, their principles and basic concepts in a range of work contexts.
Describe applications for an operating system for personal computers in a range of work contexts.

Can Do Statements:

- I can list the operators used in PHP.
- I can explain what PHP operators mean.

Ask your partner:

- Do you know how to compare two values? Explain.
- Have you ever used numeric values to perform arithmetical operations? Explain.





Let's Explore!

Kevin and Brian talk about operators.

Instructions: Read the text below.

- Karen:** I haven't studied for the next programming exam, what about you?
- Brian:** Neither have I. Is it about constants?
- Karen:** No, it's about Operators, but I didn't understand the teacher's explanation.
- Brian:** Neither did I. This teacher doesn't explain as well as Mrs. Patterson. However, if you want, we can study together. I have some information that can help us.
- Karen:** Sure, so let's start!
- Brian:** So, what do you think PHP operators mean?
- Karen:** What I understood is that these are conventions to express mathematical operations.
- Brian:** Yes! In PHP, these are conventions or symbols that permit us to perform specific functions on variables and values such as addition, multiplication, subtraction, concatenation, among others.
- Karen:** That's not as tough as I believed.
- Brian:** Indeed, it is easy.
- Karen:** Is that all, right?
- Brian:** No, we still have the types of PHP Operators.
- Karen:** Well, I learned about the Arithmetic Operators. You can use them with numeric values to carry out arithmetical operations such as addition, division, exponentiation, among others. I guess there are two or three more types of PHP operators.
- Brian:** No, including Arithmetical operators, there are seven types: Assignment, Comparison, Increment/Decrement, Logical, String, and Array operators.
- Karen:** Well, I know that the Assignment operation is in charge of using numeric values and converts them into written values. Then, the Comparison operator is used to contrast two values. Do we need to learn the function of each operator?
- Brian:** No, it's not necessary. We didn't study that in class, so it won't be included.
- Karen:** Let's hope so because learning the function of each operator is not as effortless as their general characteristics.
- Brian:** You're right. I won't study that.
- Karen:** So, neither will I.



Let's Practice!

Every operator is different.

Instructions: Read the sentences. Type in the missing words. Use: don't, neither do I, neither, haven't prepared, didn't know. Use capital letter where necessary.

Karen: I _____ the presentation for the next week, what about you?

Brian: Neither have I.

Karen: I have been reading about Logical operators, but I _____ understand them.

Brian: _____. You should make the presentation about Arithmetic operators. They are easier. For instance, I didn't know that these operators could add, subtract, and multiply.

Karen: _____ did I. So, I will make my presentation about Arithmetic operators.



Let's Practice!

Discovering the Functions of PHP Operators

Instructions: Select the correct answer.

1. Which operator is in charge of exponentiation?
 - a. Array operator
 - b. String operator
 - c. Logical operator
 - d. Arithmetic operator
2. Which operator is in charge of combining the conditional statements?
 - a. Array operator
 - b. String operator
 - c. Logical operator
 - d. Arithmetic operator
3. Which operator is in charge of comparing arrays?
 - a. Array operator
 - b. String operator
 - c. Logical operator
 - d. Arithmetic operator
4. Which operator is in charge of incrementing a variable's value?
 - a. Assignment operator
 - b. Comparison operator
 - c. Increment / decrement operator
 - d. String operator
5. Which operator is in charge of comparing two values?
 - a. Assignment operator
 - b. Comparison operator
 - c. Increment / decrement operator
 - d. String operator





Extension Activity

An extra PHP operator

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for information about Conditional Statements in PHP
- Write down your findings with the vocabulary of PHP conditional statements
- Make a presentation for your classmates and your instructor.

Self-Assessment

Yes

No

- I can investigate about more PHP operators
- I can write about more PHP operators
- I can talk about PHP operators

☐	☐
☐	☐
☐	☐



Target Words

- **concatenation** (noun): an operation of linking character strings
e.g. These are conventions or symbols that permit us to perform specific functions on variables and values such as addition, multiplication, subtraction, concatenation, among others.
- **tough** (adjective): related to difficult
e.g. That's not as tough as I believed.
- **carry out** (phrasal verb): to perform
e.g. You can use them with numeric values to carry out arithmetical operations.
- **exponentiation** (noun): algorithm used to raise an specific quantity to the power of another one
e.g: You can use them with numeric values to carry out arithmetical operations such as addition, division, among others.
- **effortless** (adjective): easy to do
e.g. Learning the function of each operator is not as effortless as their general characteristics.



Learning Points

Comparisons with As - As

We use **as – as** when we make a statement in which two things or people share a quality.

As – as is a form to make comparisons. The rule is that between **as – as** we must write an adjective or adverb.

Examples:

PHP Operators are not **as complicated as** Constants.

The Comparison operator doesn't have **as many operations as** the Arithmetical operator.

Examples

- *I didn't anticipate operators to be almost as simple as constants*
- *PHP is one of those things you don't learn by reading; you achieve it by using it. Just reading is as useful as an uncharged cellphone.*

Responses with Neither

When we want to agree to a person's negative statement, saying that we don't do or we don't like the same thing, we use **neither**.

The rule is to make this answer in the same time tense.

Examples:

Present

A: I **don't work** with PHP operators.

B: **Neither do I.**

Past

A: I **didn't know** how to use the Logical operators.

B: **Neither did I.**

Future

A: I **won't study** for tomorrow's test

B: Really? **Neither will I.**

Examples

- **Karen:** *I haven't studied for the next programming exam, what about you?*
Brian: Neither have I.
- **Brian:** *You're right. I won't study that.*
Karen: So, neither will I.

